CPSC 233 Assignments 03 Due Date: Tuesday, July 25th by 11:59 PM

Topic: Introduction to Classes (Inheritance, Constructor in inheritance, method overriding)

Collaboration

Discussing the assignment requirements with others is a reasonable thing to do, and an excellent way to learn. However, the work you hand-in must ultimately **be your work**. This is essential for you to benefit from the learning experience, and for the instructors and TAs to grade you fairly. Handing in work that is not your original work, but is represented as such, is plagiarism and academic misconduct. Penalties for academic misconduct are outlined in the university calendar.

Here are some tips to avoid plagiarism in your programming assignments.

1. Cite all sources of code that you hand-in that are not your original work. You can put the citation into comments in your program. For example, if you find and use code found on a web site, include a comment that says, for example:
   `# the following code is from https://www.quackit.com/python/tutorial/python_hello_world.cfm.`
   Use the complete URL so that the marker can check the source.
2. Citing sources avoids accusations of plagiarism and penalties for academic misconduct. However, you may still get a low grade if you submit code that is not primarily developed by yourself.
3. Discuss and share ideas with other programmers as much as you like, but make sure that when you write your code that it is your own. A good rule of thumb is to wait 20 minutes after talking with somebody before writing your code. If you find yourself exchanging code by electronic means, writing code while sitting and discussing with a fellow student, typing what you see on another person's console, then you can be sure that "your" code is not substantially your own, and your sources must then be cited to avoid plagiarism.
4. **Collaborative coding is strictly prohibited.** Your assignment submission must be strictly your code. Discussing anything beyond assignment requirements and ideas is a strictly forbidden form of collaboration. This includes sharing code, discussing code itself, or modeling code after another student's algorithm. **You cannot use (even with citation) another student's code.**
5. We will be looking for plagiarism in all code submissions, possibly using automated software designed for the task. For example, see Measures of Software Similarity (MOSS - https://theory.stanford.edu/~aiken/moss/).

Remember, if you are having trouble with an assignment, it is always better to go to your TA and/or instructor to get help, than it is to plagiarize.

Requirements:

In this assignment, you will develop a simple object oriented system. Feel free to ask, if you feel any difficulty or have any confusion.

For this assignment, you will create three classes. You can select classes as per your own. Below are the criteria of the classes

1. There should be one parent class
2. There will be two child classes.

Parent Class:

- The parent class should have min one instance variable (1 marks)
- It should have a default constructor, and a parameterized constructor. (2 marks)
- It should have getter and setter methods. (1 marks)
- It should have min one methods in addition to constructors, getters and setters. This method should perform some calculation. (2 marks)

Child Classes:

- Every child class should have two instance variables (2 marks)
- The child class should have default constructor, parametrized constructors. (2 marks)
- It should have getter and setter methods. (1 marks)
- The child class should have min one methods. This method is same in all the three classes, however each class has its own definition of the objects. This is method overriding. This method should be other than displaying just the objects itself. (3 marks)

Main method:

In your main function, you should demonstrate the following functionalities

1. Create objects of parent class and child classes
2. Demonstrate how method overriding work. You can call same methods with different class objects. (3 marks)

UML:

Also submit a UML class diagram for your system. Identify the relationship. Hand draw UML diagram is acceptable. (3 marks)

Marks will be deducted if parent methods/constructors are not called appropriately from the child classes.

Submission:

Use the University of Calgary Desire2Learn system (http://d2l.ucalgary.ca) to submit your assignment work online. Log in using your UofC eID and password, then find our course, cpsc 233, in the list. Then

navigate to *Assessments -> Dropbox Folders*, and find the folder for Assignment #3 here. You should submit the code file in D2L.

Grading Rubric:

There are total 20 marks for this assignment. See the description for the assignment.